MILITARY SPECIFICATION

FUZE MOVEMENT ASSEMBLY 10522353

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Movement Assembly 10522353.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes: General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Test For
MIL-STD-1167	- Ammunition Data Card
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

Armament Command

F10522353	- Movement Assembly
D9282877	- Packing and Marking for Box, Wood Ammunition Packing for Movement Assembly P/N 10522353

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2.1 DRAWINGS (Cont'd)

D9282876	- Packing and Marking for Box Fiberboard (Outer) Ammunition Packing for Movement Assembly P/N 10522353
IEL 10522353	- Inspection Equipment List for Movement Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications except as provided in 6.1.

3.2 Dynamic regulation.-The movement subassembly shall be regulated as specified on the applicable drawing. The timing error between 2 and 100 seconds shall not exceed 0.18% while spinning at the specified revolutions per minute (rpm).

3.3 Environmental - (with the movement assembly installed in Fuze, MTSQ, M564)

3.3.1 Jolt.-The movement assembly shall comply with MIL-STD-331, Test No. 101.1, except as specified herein.

3.3.2 Jumble.-The movement assembly shall comply with MIL-STD-331, Test No. 102.1.

3.3.3 Transportation - vibration.-The movement assembly shall comply with MIL-STD-331, Test No. 104.

3.4 Ballistics - (with movement assembly installed in Fuze, MTSQ, M564) (See 6.2)

3.4.1 The movement assembly shall not function prematurely (See 6.2.1).

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3.4.2 Time-to-burst (ambient temperature).-The movement assembly shall initiate detonation of the round during flight within the time limits specified in Table I, when fired as follows:

Weapon	Charge	Setting
155mm Howitzer	1 (680 fps)	3 seconds
105mm Howitzer	7 (1550 fps)	60 seconds

TABLE I

Weapon	Charge	Fuze Setting	Individual Round Function Time Limits (seconds)	
155mm How.	1	3 seconds	LL = 2.88	UL = 3.47
105mm How.	7	60 seconds	LL = 57.70	UL = 59.60

3.4.3 Excess pressure.-The movement assembly shall initiate detonation of the round during flight when fired in the 105mm Howitzer at 112% of the normal charge 7 pressure with the fuze set at 60 seconds.

3.4.4 Low temperature.-The movement assembly with fuze temperature conditioned at minus 65 to minus 70 degrees Fahrenheit shall initiate detonation of the round during flight when fired as specified in 3.4.2 (See 6.3).

3.4.5 High temperature.-The movement assembly with fuze temperature conditioned at plus 145 to plus 150 degrees Fahrenheit shall initiate detonation of the round during flight when fired as specified in 3.4.2 (See 6.3).

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions or any similar defect.

3.5.2 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.5.3 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.4 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in process and cleaning shall not adversely affect the finish or contaminate the parts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of movement assemblies shall be comprised of components each produced by one manufacturer, by one process in accordance with the same drawing and revision and specification and revisions, as applicable, and assembled on one assembly line.

4.2 First article sample.-When required, the first article sample shall be as specified in the contract and shall consist of the following:

240 (aggregate) component parts as delineated in 4.2.2
350 (aggregate) subassemblies as delineated in 4.2.3
98 (aggregate) fuze assemblies (Dwg. F10534285) as delineated in 4.2.5

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, 85 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Two-hundred and forty (240) component parts, 350 subassemblies and 13 fuze assemblies shall be tested, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each of the 12 components shall be tested as in 4.3.3.1 (Hardness), 4.3.3.2 (Load) and 4.3.3.4 (Bend).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each of the 12 subassemblies shall be tested as in 4.3.3.5.1 (Static load).

4.2.3.2 Twenty-five (25) of each of the 2 subassemblies shall be tested as in 4.3.3.5.2 (Torque).

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4.2.4 Acceptance criteria.-Failure of any component or subassembly to comply with the requirements of the applicable drawing for the above characteristics shall be cause for rejection of the First Article sample.

4.2.5 Fuze assemblies.

4.2.5.1 MIL-STD tests.-Thirteen (13) fuze assemblies shall be subjected to the following MIL-STD tests, with the fuzes set on safe "S".

Test	Quantity	Requirement	Test Method
Jolt	4	3.3.1	4.4.8.1
Jumble	4	3.3.2	4.4.8.2
Transportation-Vibration	5	3.3.3	4.4.8.3

4.2.5.1.1 Acceptance criteria.-The criteria of the applicable MIL-STD shall apply as determined by disassembly and examination in accordance with the procedures in 4.4.8. Where the accept/reject decision is based on engineering judgement, concurrence by the Technical Agency must be obtained.

4.2.5.2 Ballistics.-Eighty-five (85) fuzes of the First Article Sample shall be subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Time-to-Burst (Ambient Temperature)			
155mm Howitzer Charge 1	20	3.4.1 & 2	4.4.9.1
105mm Howitzer Charge 7	20	3.4.1 & 2	4.4.9.1
Excess Pressure (Ambient Temperature)	25	3.4.1 & 3	4.4.9.2
Low Temperature (-65°to -70°F)			
155mm Howitzer Charge 1	5	3.4.1 & 4	4.4.9.3
105mm Howitzer Charge 7	5	3.4.1 & 4	4.4.9.3
High Temperature (+145°to +150°F)			
155mm Howitzer Charge 1	5	3.4.1 & 5	4.4.9.3
155mm Howitzer Charge 7	5	3.4.1 & 5	4.4.9.3

4.2.5.2.1 Acceptance criteria.-The occurence of any of the following shall be cause for rejection of the First Article Sample.

a. A premature burst in any phase (See 6.2.1)

b. Any of the following in the ambient Time-to-Burst Phase at either time setting:

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4.2.5.2.1 (Cont'd)

(1) More than one time failure (See 6.2.2).

(2) More than one outlier-offtime (See 6.2.3).

(3) Failure to meet the time limits in Table I by an estimated lot percent defective greater than the appropriate value given in Table II (See 4.4.9.1.1).

c. More than two time failures (6.2.2) in the Excess Pressure Phase.

d. More than two time failures (6.2.2) in either of the Environmental Phases (Low Temperature, High Temperature).

TABLE II

3 Sec. Setting

Sample Size	Allowable Percent Defective
20	2.95
19	2.96
18	2.97
17	2.99
16	3.01
15	3.05

60 Sec. Setting

20	4.00
19	4.01
18	4.03
17	4.06
16	4.12
15	4.20

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

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4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect basis in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each major defect listed. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.1 Plate No. 1 First Lamina Drawing F10523181 covering a detail of Drawing F10522353.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred or damaged	Visual
105.	Chamfer on holes M and 20 missing or incorrect	Visual

4.3.2.2 Plate No. 2 Second Lamina Drawing F8595968, covering a detail of Drawing F10522353.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Diameter of holes M and 20, incorrect	Gage
102.	Location of holes M and 20, incorrect	Gage
103.	Surface finish of holes M and 20, improper	Visual
104.	Metal burred or damaged	Visual
105.	Chamfer on holes M and 20 missing or incorrect	Visual

4.3.2.3 Plate No. 2 First Lamina Drawing F8595967, covering a detail of Drawing F10522353.

CRITICAL		
	None defined	
MAJOR	AQL = 1.0%	
101.	Flatness incorrect	Gage
102.	Metal burred or damaged	Visual

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4.3.2.4 Slide, Pallet Lock Dwg. C10521169, covering a detail of Dwg. F10522353.

Defect	Method of Inspection
CRITICAL	
1. Large radius on extension above max.	Gage
MAJOR	
None defined	

4.3.2.5 Gear, Escapement Dwg. D10521109, covering a detail of Dwg. F10522353.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

4.3.2.6 Escapement Gear and Pinion Assembly Drawing C10521112, covering a detail of Drawing F10522353.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Eccentricity of gear	Gage
102. Perpendicularity of gear, incorrect	Gage
103. Burr	Visual

4.3.2.7 Escapement Lever Assembly Drawing F8593015, covering a detail of Drawing F10522353.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Spring burred or damaged	Visual
102. Improper assembly	Visual

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4.3.2.8 Arbor, Escapement Dwg. D10534857, covering a detail of Dwg. F10522353.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Perpendicularity of lever seat, incorrect	Gage
102. Diameter of pivots, incorrect	Gage
103. Metal burred or damaged	Visual

4.3.2.9 Firing Pin, Dwg. C10521154, covering a detail of Dwg. F10522353.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of striker point, improper	Gage
102. Surface finish on shoulder angle, improper	Visual

4.3.2.10 Firing Arm Assembly, Drawing C10521115, covering a detail of Drawing F10522353.

CRITICAL	
1. Fracture of upright bend of firing arm	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 1.0%	
101. Insecure weight	Manual/Visual
102. Improper or incomplete assembly	Visual

4.3.2.11 Timing Disc Assembly, Drawing D10520268, covering a detail of Drawing F10522353.

CRITICAL	
1. Safety disc tabs not engaged with timing disc bushing slots	Visual
MAJOR AQL = 0.40%	
101. Adjusting segment radius extends beyond periphery of timing disc	Visual
102. Safety disc radius below periphery of timing disc	Visual
103. Improper or incomplete assembly	Visual

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4.3.2.12 Movement Subassembly (after dynamic regulation), Drawing F10522354, covering a detail of Drawing F10522353.

Defect	Method of Inspection
CRITICAL	
1. Firing pin fires on "not fire" position of firing arm	Gage
2. Extension of firing arm upright above top plate less than minimum (with firing arm shaft shoulder bottomed on plate No. 2)	Gage
MAJOR AQL = 0.40%	
101. Firing pin does not fire on "must fire" position of firing arm	Gage
102. Firing pin protrusion less than min. in fired position	Gage
103. Extension of firing arm upright above top plate exceeds maximum (with firing arm shaft shouldered against top plate)	Gage
104. Firing arm shaft spring protrudes	Visual
105. Arbor centerhole missing	Visual
106. Burrs and irregularities which prevent proper seating of movement in the fuze body	Visual
107. Firing pin protrudes	Manual/Visual
108. Staking of body assembly screws inadequate (check prior to dynamic regulation)	Visual

4.3.2.13 Movement Assembly, Drawing F10522353.

CRITICAL	
1. Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual/Visual
2. Timing disc retainer insecure, damaged or missing	Manual/Visual
3. Distance from top of top plate to top of retainer exceeds maximum (after staking of retainer)	Gage
4. Escapement gear not locked in sufficiently by lever pallet	Visual
MAJOR AQL = 0.40%	
101. Total height exceeds max.	Gage
102. Improper or incomplete assembly	Manual/Visual

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4.3.2.13 (Cont'd)

Defect	Method of Inspection
103. Escapement lever not locked on proper side of pallet lock slides	Visual
104. Foreign matter	Visual
105. Firing pin striker point damaged	Visual

4.3.2.14 Packaging, packing and marking.-The sample size and acceptance criteria for packaging, packing and marking shall be in accordance with MIL-P-60412.

4.3.2.15 Process control audit.-Periodic (at least once during each 8-hour shift) unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawing. Bending adjustment of the firing arm upright is not permitted at any time.

a. Laminated Plates No. 1 and No. 2 Assembly, Drawing F8595991

(1) Application of lubricant

b. Movement Subassembly, Drawing F10522354

(1) Application of lubricant

(2) Mainspring wound before assembly of screw

(3) Dynamic regulation operation

(4) Movement fully wound after dynamic regulation

(5) Backlash removal operation after final wind

(6) Top and bottom assembly screws staked prior to Dynamic Regulation

c. Movement Assembly, Drawing F10522353

(1) Locknut torqued prior to timing disc torque test

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (Destructive).-Twenty (20) samples from each lot of the following components shall be subjected to the hardness test. Where plating is required, the sample parts tested shall be unplated, where feasible. Failure of any component to comply with the requirements specified on the applicable drawing shall be casue for rejection of the lot of components, as applicable (see 4.4.1).

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4.3.3.1 (Cont'd)

Component	Drawing Number
Pin, Firing	C10521154
Gear, Escapement	D10521109
Pinion, Escapement	D10521101
Pinion, No. 1	D10521119
Pinion, No. 2	D10521120
Pinion, No. 3	D10521118
Pinion, No. 4	C10521123
Arm, Firing (see 4.3.3.4)	D10521116
Guide, Adjusting	C8594273
Lever, Escapement	F8593023
Spring, Timing Disc (see 4.3.3.2)	C10521161
Washer, Timing Disc	B8594267

4.3.3.2 Load - Timing Disc Spring (Destructive).-The same 20 timing disc springs subjected to the hardness test shall be subjected to the load specified on Drawing C10521161. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of timing disc springs (see 4.4.2).

4.3.3.3 Compression - Spring (Destructive).-Twenty firing pin springs from each lot of firing pin springs shall be subjected to the compression specified on Drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of firing pin springs (see 4.4.3).

4.3.3.4 Bend - Firing Arms (Destructive).-The same 20 firing arms that comply with the hardness requirement shall be subjected to the bend specified on Drawing D10521116. Failure of any firing arm to comply with the requirement shall be cause for rejection of the lot of firing arms (see 4.4.4).

4.3.3.5 Static load and torque.

4.3.3.5.1 Static load.-Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.5.1).

Static Load	Drawing Number
Mainspring Housing Assembly (nondestructive)	F10520256
Disc and Stud Assembly	C8593103
Escapement Gear and Pinion Assembly	C10521112
Escapement Lever Assembly	F8593015

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4.3.3.5.1 (Cont'd)

Static Load	Drawing Number
Firing Arm Assembly (Test Pin and Arm)	C10521115
No. 1 Gear and Pinion Assembly	D10521102
No. 2 Gear and Pinion Assembly	C10521104
No. 3 Gear and Pinion Assembly	C10521106
No. 4 Gear and Pinion Assembly	C10521111
No. 5 Gear and Arbor Assembly	C10522355
Pallet Lock Assembly	B10521168
Lever Assembly	B10521148

4.3.3.5.2 Torque.-Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.5.2).

Torque	Drawing Number
Escapement Lever Assembly	F8593015
No. 5 Gear and Arbor Assembly	C10522355

4.3.3.6 Movement subassembly.

4.3.3.6.1 Kinetic energy of firing pin (Nondestructive).-Fifty (50) movement subassemblies from each lot shall be tested to determine compliance with the requirements specified on Drawing F10522354. Failure of any assembly to comply with the requirement shall be cause for rejection of the lot of movement subassemblies (see 4.4.6.1).

4.3.3.6.2 Spin function (Nondestructive).-The pallet lock and safety lever assemblies within each movement subassembly produced shall be tested to determine compliance with the function requirement of Drawing F10522354. Failure to comply with the requirement shall be cause for rejection of the subassembly (see 4.4.6.2).

4.3.3.6.3 Dynamic regulation (Nondestructive).-The regulation tape record (100 seconds) of each movement subassembly produced shall be inspected to determine acceptability of the movement. Any movement subassembly which fails to comply with the requirement of 3.2 shall be rejected (see 4.4.6.3).

4.3.3.6.4 Spin nonfunction (Nondestructive).-Each movement subassembly produced shall be tested to determine compliance with the nonfunction requirement of Drawing F10522354. Failure of the lever lock slides to remain in the locked position shall be cause for rejection of the subassembly (see 4.4.6.4).

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4.3.3.7 Timing disc torque (Nondestructive).-The timing disc of each movement assembly produced shall be subjected to the torque specified on Drawing F10522353. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.7).

4.3.3.8 Environmental tests (Lots) (Destructive).-Twenty-two fuze movement assemblies from each production lot shall be subjected to the following MIL-STD tests, with the fuzes set safe "S".

Test	Quantity	Requirement	Test Method
Jolt	4	3.3.1	4.4.8.1
Jumble	4	3.3.2	4.4.8.2
Transportation-Vibration	5	3.3.3	4.4.8.3

4.3.3.8.1 Acceptance criteria.-The criteria of the applicable MIL-STD shall apply as determined by disassembly examination and tests in accordance with the procedures in 4.4.8. Where the accept/reject decision is based on engineering judgement, concurrence by the responsible technical agency must be obtained.

4.3.3.9 Ballistics (Destructive).-Forty movement assemblies from each lot shall be subjected to the following ballistic test phases:

Phases (Ambient Temp.)	Quantity	Requirement	Test Method
Time-to-Burst			
155mm Howitzer Charge 1	20	3.4.1 & 2	4.4.9.1
105mm Howitzer Charge 7	20	3.4.1 & 2	4.4.9.1

4.3.3.9.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature burst in any phase (see 6.2.1)

b. Any of the following in the Ambient Time-to-Burst Phase at either time setting.

(1) More than one time failure (6.2.2)

(2) More than one outlier-offtime (6.2.3)

(3) Failure to meet the time limit specified in Table I by an estimated lot percent defective greater than the appropriate value given in Table II (see 4.4.9.1.1).

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed in IEL-10522353. Equipment design shall be in accordance with the code designation on the applicable IEL.

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4.4 Test methods and procedures.

4.4.1 Hardness.-The test samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual value.

4.4.2 Load (Timing Disc Spring).-The test shall be performed using the values as specified on Drawing C10521161. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.3 Compression.-The firing pin springs shall be compressed for the length of time specified on Drawing C10521132. The springs that have been subjected to the compression test shall be scrapped.

4.4.4 Bend - Firing Arm.-The test shall be performed as specified on Drawing D10521116. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.5 Static load and torque.

4.4.5.1 Static load.-Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicable drawings. The load when applied to pins shall be in the opposite direction of assembly of the pins. All samples tested shall be scrapped, except the mainspring housing assemblies.

4.4.5.2 Torque.-Escapement lever assemblies shall be tested prior to insertion of the escapement spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

4.4.6 Movement Subassembly.

4.4.6.1 Kinetic energy of firing pin.-The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position.

4.4.6.2 Spin function.-The spin function test shall be performed on wound subassemblies. The subassemblies shall be rotated at the rpm specified on Drawing F10522354 in a clockwise direction when viewed from the bottom, for a period of not less than 10 seconds at constant rpm. The movement shall begin functioning immediately at the specified rpm and the safety lever shall be released during the spin. Compliance with the requirement shall be determined after the test by the position of the safety lever (contacting the periphery of the collar). At the conclusion of the test, each mechanism shall be rewound. The function test shall be performed prior to the nonfunction tests.

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4.4.6.3 Dynamic regulation monitoring.-The following procedure shall be observed: The movement subassemblies shall be regulated, using the operational procedures furnished with the equipment. The movement subassemblies shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6,000 ± 15 rpm. Timekeeping and performance shall be checked over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run.

4.4.6.4 Spin nonfunction.-The spin nonfunction test shall be performed on completely wound subassemblies (see 6.5). The subassemblies shall be rotated at the rpm specified on Drawing F10522354 in a clockwise direction when viewed from the bottom for a period of not less than 3 seconds at constant rpm to determine compliance with the requirement (Neither lever lock slide releases the escapement lever). This test shall be the last spin test to which the movement is subjected.

4.4.7 Timing disc torque.-The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest readings will be taken in one uninterruped complete revolution of the timing disc simulating setting pin and timing disc lug engagement for determination of compliance with the requirments of Drawing F10522353.

4.4.8 Environmental test procedures.-The component parts of the movement assembly which have been disassembled and inspected following tests, shall be scrapped. Where spin testing is required, the nonarming test shall be performed prior to the arming tests.

4.4.8.1 Jolt.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101.1 except that two arms loaded with fuzes and dummies may be used for the test. The fuze shall be disassembled and inspected as follows:

a. Remove head assembly and check movement safety disc relative to the firing arm.

b. Remove lower cap and housing. Check timing disc retainer.

c. Remove movement assembly from body. Check position of firing pin, pallet lock slides.

d. Disassemble movement assembly to the extent that all critical parts can be examined, i.e., all gears, pinions, etc., for broken pivots, teeth, or timing disc spring collapse.

4.4.8.2 Jumble.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102.1. The fuze shall be disassembled and inspected as outlined in 4.4.8.1.

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4.4.8.3 Transportation-Vibration.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104, as follows:

a. Parallel to longitudinal axis, nose up.

b. First transverse axis to be parallel to a line running through the 100 second marking.

c. Second transverse axis to be parallel to a line running through the 70 second marking. The fuzes subjected to this test shall be disassembled and inspected as follows:

(1) Remove head assembly and check movement safety disc relative to the firing arm.

(2) Remove lower cap and housing. Check timing disc retainer.

(3) Remove movement assembly from the body. Check position of firing pin. Spin for nonfunction and function requirements.

(4) Disassemble movement assembly to the extent that all critical parts can be examined, i.e., all gears and pinions, etc., for broken pivots, teeth, or timing disc spring collapse.

4.4.9 Ballistics.-The complete fuze assembly, Drawing F10534285 shall be assembled to a suitable projectile and fired as follows:

4.4.9.1 Time-to-burst phase.-The sample fuzes shall be fired as specified in 3.4.2 under the conditions delineated in 4.2.5.2 or 4.3.3.9 as appropriate. Time-to-burst shall be measured to the nearest millisecond by using two independent fuze chronograph systems. Time-to-burst shall also be measured by using two electric clocks or stop watches. All measurements shall be recorded. (see 6.2.3.2)

4.4.9.1.1 Lot percent defective.-The time values obtained by the chronograph systems shall be used to calculate the estimate of lot percent defective. Where both chronograph systems provide a time value, the average shall be taken as the individual fuze value. If one chronograph system fails to provide a time value, the one time value obtained from the functioning chronograph system shall be used as the individual fuze value. Individual fuze values shall be rounded off to the second decimal place using the method prescribed by the American Society for Testing Materials. The estimate of lot percent defective shall be calculated using the method prescribed in MIL-STD-414, Example B3. For sample sizes 16 through 19 the estimated lot percent defective value shall be determined by interpolation between the values given in Table B5, MIL-STD-414 for sample sizes 15 and 20. All appropriate time values obtained for each time-to-burst phase shall be used in the initial calculation of the sample average ($\bar{x}$) and standard deviation (s). The standard deviation value obtained shall then be used to determine statistical outliers (non-normal values) by using the criteria $\bar{x}$ plus or minus 2.88 times the standard

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4.4.9.1.1 (Cont'd)

deviation. A value judged to be an outlier shall be excluded and the remaining values used to recalculate $\bar{x}$ and s. A minimum of 15 statistically normal values are required for each time-to-burst phase. (see 6.2.3.1)

4.4.9.2 Excess pressure phase.-The sample movement assemblies shall be fired as specified in 3.4.3 under the condition delineated in 4.2.5.2. Time-to-burst shall be measured in the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.9.3 Extreme temperature phase.-The sample movement assemblies shall be fired as specified in 3.4.4 and 3.4.5 under the conditions delineated in 4.2.5.2 and 4.4.9.3.2. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.9.3.1 Temperature conditioning (-65°F and +145°F).-The test chamber shall be maintained at a temperature of minus 65°F or plus 145°F as applicable, until stabilization of the fuze temperature is reached (see 6.3). After stabilization, the fuze shall be saturated at minus 65°F or plus 145°F, as applicable, for one hour, minimum.

4.4.9.3.2 High and low-temperature firings.-For the high and low temperature ballistic firings only the fuze shall be temperature conditioned. The elapsed time from the conditioning chamber to the firing in the weapon shall not exceed five (5) minutes. Reconditioning is not permitted.

5. PREPARATION FOR DELIVERY

5.1 Preservation packing and marking.-Preservation, packaging, packing and marking shall be as specified in the contract and MIL-P-60412, and Drawing D9282877 (Level A) and Drawing D9282876 (Level B).

6. NOTES

6.1 Dimensions.

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specification and drawings is mandatory. The contractor will not assume, nor does the Government guarantee, that all possible combinations permitted by the tolerance limits of the specification and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

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6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate movement assembly performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on movement assembly functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the movement assembly.

6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Ballistic parameters.

6.2.1 Premature burst.-Shall be defined as the occurrence of either of the following:

a. The round detonates in the bore of the gun.
b. The round detonates before 45 seconds when set at 60 seconds or before 1.9 seconds when set at 3 seconds.

6.2.2 Time failure.-A time failure is considered to have occurred if the round fails to detonate during flight.

6.2.3 Outlier-offtime.-An outlier-offtime is a statistical outlier (Non-normal value) which falls beyond the appropriate time limits specified in Table 1.

6.2.3.1 Where the number of statistically normal time values obtained in either time-to-burst phase is less than 15 and the lot is not rejected because of other reasons, the disposition of the lot shall be referred to the responsible procuring activity. The Government reserves the right to require the contractor to submit additional test samples. The number of additional test samples shall be sufficient to obtain a combined total of 20 values in order that the occurrence of outliers will not reduce the number of normal values to less than the minimum required. All normal values obtained in the combined tests shall be used to determine lot acceptance or rejection. Attribute characteristic failure, e.g., duds, occurring in the additional samples shall not be used to reject the lot.

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6.2.3.2 Electric clock and stop watch time values.-Where only electric clock or stop watch time measurements are obtained and these values indicate a premature burst or outlier-offtime which affects lot disposition, the lot shall be referred to the responsible procuring activity.

6.2.3.3 Defective fuzes.-Where sample fuzes are found to be defective as received at the ballistic test facility, all tests of the lot sample shall be held in abeyance pending investigation by the responsible procuring activity.

6.2.4 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.3 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated fuze may be used for this purpose. (see 4.4.9.3.2)

6.4 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6.5 Completely wound movement.-A completely wound movement is one which has been wound to the stop and had gear train blacklash removed in accordance with the drawing.

6.6 Statement of Safety Requirements.

6.6.1 Safety Limitations and Precautions Relating to Production Equipment and Facilities:

Assembly of explosive elements into fuzes shall be performed under protective hoods with static electricity controls. Protective barrier should be used when performing the MIL-Standard tests.

6.6.2 Hazards Presented by Materials:

Possible injury (burns or dismemberment) to limbs and body from flash or detonation of explosive elements.

6.6.3 Known Hazards in Hazardous Areas:

All areas in which explosive elements are used in assembly operations or stored.

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6.6.4 Specific Instructions Relative to Manufacture, Handling, Storage and Shipment of Material: The number of persons exposed to explosive operations and the quantities of explosive materials should at all times be the minimum consistent with safety and the efficiency of operation. Signs setting forth personnel and explosive limits also should be posted. Prior to starting operations including any of the hazardous materials or items described herein, a standard operating procedure should be developed and approved by the contractor outlining the specific requirements to safely perform the particular operation. All personnel concerned should be cognizant of SOP contents. Applicable portions of the approved procedure should be conspicuously posted convenient to all stations involved in the operation for guidance of all personnel.

6.6.5 Explosive Hazard Classification of Materials:

Assembled fuzes - Class 1, Detonators and closing cup (lead charge) assemblies in Storage - Class 7.

6.7 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1390-A135

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